17 Lecture 17 (February 24, 2025, est.): The Verlinde category

Scribe: Guillermo Sanmarco

Throughout this lecture, k is algebraically closed of characteristic $p > 0$. We apply semisimplification to representations of the cyclic group C_p. Recall that an indecomposable survives semisimplification precisely when its categorical dimension is nonzero, provided traces of nilpotent endomorphisms vanish. This hypothesis holds in representation categories, where the categorical trace is the ordinary linear trace.

17.1 Representations of the cyclic group

Choose a generator σ of C_p. Writing $t = \sigma - 1$ gives

$$k[C_p] = k[\sigma]/(\sigma^p - 1) \cong k[t]/(t^p).$$

Thus a representation is a vector space with a nilpotent endomorphism whose pth power is zero. Jordan normal form gives exactly p indecomposable representations:

$$J_n = k[t]/(t^n), \qquad 1 \le n \le p.$$

In particular, J_1 is the trivial representation and J_p is the regular representation.

17.1 Definition. The *Verlinde category* is the symmetric tensor category

$$\mathrm{Ver}_p = \overline{\mathrm{Rep}_k(C_p)}.$$

Write L_n for the image of J_n in this semisimplification. Its simple objects are $L_1, \dots, L_{p-1}$, with $L_1 = \mathbb{1}$, whereas $L_p = 0$. Their categorical dimensions are $\dim(L_n) = n \cdot 1_k$.

There are finitely many simples here, so Ver_p is a symmetric fusion category. Finiteness of the original group alone would not ensure this conclusion for its semisimplification: representation categories of other finite groups can have infinitely many surviving indecomposables. See [EO19] for examples and the general construction.

17.2 Tensor products of Jordan blocks

17.2 Proposition. *The J_n are self-dual, and*

$$J_2 \otimes J_n \cong \begin{cases} J_2, & n = 1, \\ J_{n-1} \oplus J_{n+1}, & 2 \le n \le p-1, \\ J_p^{\oplus 2}, & n = p. \end{cases}$$

For $1 \le n \le p-1$ one also has

$$J_{p-1} \otimes J_n \cong J_{p-n} \oplus J_p^{\oplus(n-1)}.$$

Proof. On a dual representation, σ acts by the inverse transpose. The inverse of a unipotent Jordan block is again a unipotent matrix with one Jordan block of the same size, proving self-duality.

For the first tensor-product formula, identify the underlying vector space with $k[x,y]/(x^2,y^n)$. The generator acts by multiplication by $(1+x)(1+y)$. Set $u = x(1+y)$; this is an invertible change of variable, so the algebra is also $k[u,y]/(u^2,y^n)$, and t acts by multiplication by $u+y$. Hence

$$t^r = y^r + ruy^{r-1}.$$

For $2 \le n < p$, the nilpotency index is $n+1$, since $t^n = nuy^{n-1} \neq 0$. Moreover, the kernel of multiplication by t has dimension two. Indeed, $t(A(y)+uB(y)) = 0$ means

$$yA = 0, \qquad A + yB = 0,$$

so $A = ay^{n-1}$ and $B = -ay^{n-2} + by^{n-1}$. There are therefore two Jordan blocks; their sizes are $n+1$ and $n-1$ because the total dimension is $2n$. If $n = p$, the same kernel computation gives two blocks, each of size at most p, and their total size is $2p$. The case $n = 1$ is immediate.

Finally, tensor the exact sequence

$$0 \longrightarrow J_{p-1} \longrightarrow J_p \longrightarrow J_1 \longrightarrow 0$$

with J_n. The middle term is $J_p^{\oplus n}$: for any finite group, the regular representation tensored diagonally with V is isomorphic to $\dim(V)$ copies of the regular representation, by the map $g \otimes v \mapsto g \otimes g^{-1}v$. A surjection $(k[t]/t^p)^n \to k[t]/t^n$ can be changed by an invertible basis transformation of its source to the standard quotient on its first factor and zero on the others. To see this, one of the images of the standard generators must have nonzero constant term; make it the first generator, multiply it by a unit to make its image 1, and subtract suitable multiples of it from the remaining generators. The kernel is then $t^n(k[t]/t^p) \oplus (k[t]/t^p)^{n-1}$, as claimed. ∎

17.3 Theorem (Fusion rules)**.** *For $1 \le i,j \le p-1$,*

$$L_i \otimes L_j \cong \bigoplus_{r=1}^{\min(i,j,p-i,p-j)} L_{|i-j|+2r-1}.$$

In particular,

$$L_2 \otimes L_n = L_{n-1} \oplus L_{n+1}, \qquad L_{p-1} \otimes L_n = L_{p-n},$$

where $L_0 = L_p = 0$ in the first formula.

Proof. The two displayed special cases follow from the proposition. For $p > 2$, write $x = [L_2]$ in the Grothendieck ring and define polynomials $f_0 = 0$, $f_1 = 1$, $f_{n+1} = xf_n - f_{n-1}$. The recurrence gives $[L_n] = f_n(x)$ and $f_p(x) = 0$. Induction using the same recurrence gives the polynomial identity

$$f_i(x)f_j(x) = \sum_{r=1}^{\min(i,j)} f_{|i-j|+2r-1}(x).$$

Modulo $f_p(x)$ one has $f_{p+a}(x) = -f_{p-a}(x)$ for $a \geq 0$ in the range in question. The terms above p consequently cancel the reflected terms below p, and the term f_p vanishes. The remaining indices start at $|i-j|+1$ and end at $\min(i+j-1, 2p-i-j-1)$, giving the formula. Since the category is semisimple, equality of these classes is an isomorphism of objects. For $p = 2$ there is only the simple unit, and the formula is immediate. ∎

These are the truncated Clebsch–Gordan rules also occurring in the semisimplified quantum $\mathfrak{sl}_2$ category at a primitive $2p$th root of unity in characteristic zero. This comparison concerns the fusion rules; the categories have different ground-field characteristics. For the positive-characteristic construction and fusion rules, see [Ost15, Section 3].

17.3 Small characteristics and failure of fiber functors

For $p = 2$, the only surviving simple is the unit, so $\mathrm{Ver}_2 \simeq \mathrm{Vect}_k$. For $p = 3$, the simples are L_1 and L_2, with $L_2^{\otimes 2} = \mathbb{1}$. Its self-symmetry is -1: in $J_2 \otimes J_2$, the alternating line is a copy of J_1 on which the flip is -1, whereas the symmetric square is J_3 and disappears. Thus $\mathrm{Ver}_3 \simeq \mathrm{SVec}_k$.

For $p = 5$, the fusion rule gives

$$L_3 \otimes L_3 \cong \mathbb{1} \oplus L_3.$$

If there were a symmetric fiber functor to SVec_k, the total vector space dimension n of the image of L_3 would be a positive integer satisfying $n^2 = n + 1$, which is impossible. Here one must use an ordinary nonnegative integer dimension, not the categorical dimension in k: the categorical dimension $3 \in k$ does satisfy this equation in characteristic 5.

17.4 Proposition. *For every prime $p \geq 5$, Ver_p has no fiber functor to Vect_k or to SVec_k.*

Proof. Suppose such a functor exists, and let d_i be the total dimension of the image of L_i. Faithfulness gives positive integers d_i for $1 \leq i \leq p-1$. Set $d_0 = d_p = 0$, $d_1 = 1$, and $d = d_2$. The fusion rules imply

$$d d_i = d_{i-1} + d_{i+1} \quad (1 \leq i \leq p-1).$$

Since $d^2 = 1 + d_3 \geq 2$, one has $d \geq 2$. Starting with $d_1 - d_0 = 1$, the recurrence gives

$$d_{i+1} - d_i = (d-2)d_i + (d_i - d_{i-1}) \geq 1$$

by induction, including the last step $i = p-1$. Thus $d_p \geq p$, contrary to $d_p = 0$. ∎

17.5 Example (A characteristic-two obstruction)**.** There are also symmetric tensor categories in characteristic 2 with no fiber functor to Vect_k, even though Ver_2 itself is Vect_k. Consider the Hopf algebra

$$H = k[d]/(d^2), \qquad \Delta(d) = d \otimes 1 + 1 \otimes d,$$

with triangular structure $R = 1 \otimes 1 + d \otimes d$. The identities $(\Delta \otimes \mathrm{id})(R) = R_{13}R_{23}$ and $(\mathrm{id} \otimes \Delta)(R) = R_{13}R_{12}$ follow by expanding and using $d^2 = 0$; also $R_{21}R = 1$ because $2 = 0$. Thus the symmetry on $\mathrm{Rep}(H)$ is $c = \tau \circ R$, where τ is the usual flip.

Let X have basis e, f with $de = 0$ and $df = e$. Then

$$c(e \otimes e) = e \otimes e, \qquad c(e \otimes f) = f \otimes e, \qquad c(f \otimes f) = f \otimes f + e \otimes e.$$

The image of $\mathrm{id} - c$ is spanned by $e \otimes e$ and $e \otimes f + f \otimes e$. Consequently $\mathrm{Sym}^2 X = \mathrm{coker}(\mathrm{id} - c)$ is two-dimensional, with basis the classes of $e \otimes f$ and $f \otimes f$. The action of d on both classes is zero, so $\mathrm{Sym}^2 X \cong \mathbb{1}^{\oplus 2}$. On the other hand, X is an extension of $\mathbb{1}$ by $\mathbb{1}$. An exact symmetric fiber functor F to Vect_k would give $\dim F(X) = 2$ and preserve this cokernel. This would imply both $\dim F(\mathrm{Sym}^2 X) = 2$ and $\dim \mathrm{Sym}^2(F(X)) = 3$, a contradiction.

17.4 Tensor subcategories of the Verlinde category

Assume now that $p \geq 5$. Let Ver_p^+ be the full semisimple subcategory with simples

$$L_1, L_3, L_5, \ldots, L_{p-2}.$$

The fusion rules show that it is closed under tensor products and duals, and that it is generated by L_3. More explicitly, $L_3 \otimes L_{2r+1}$ contains L_{2r+3} whenever $3 \leq 2r + 3 \leq p - 2$, starting with L_3 itself.

The object $\varepsilon = L_{p-1}$ is invertible of order two, with $\dim(\varepsilon) = -1$. For an invertible object in a symmetric category, the self-symmetry is its categorical dimension times the identity (compute the trace using its inverse object). Thus the subcategory generated by ε is SVec_k, with ε corresponding to the odd line.

17.6 Proposition. *The full tensor subcategories of* Ver_p *are exactly*

$$\mathrm{Vect}_k, \qquad \mathrm{SVec}_k, \qquad \mathrm{Ver}_p^+, \qquad \mathrm{Ver}_p\,.$$

Moreover, multiplication induces a symmetric equivalence

$$\mathrm{SVec}_k \boxtimes \mathrm{Ver}_p^+ \xrightarrow{\sim} \mathrm{Ver}_p\,.$$

Here a full tensor subcategory is understood to be closed under finite direct sums and subobjects, and $\boxtimes$ *denotes the product whose simple objects are pairs of simple objects of its factors.*

Proof. If a tensor subcategory contains L_i with $2 \leq i \leq p - 2$, then $L_i \otimes L_i$ contains L_3, so the subcategory contains Ver_p^+. If i is even, then $p - i$ is odd, hence L_{p-i} already belongs to Ver_p^+. The product $L_i \otimes L_{p-i}$ contains $L_{p-1} = \varepsilon$, so the subcategory is all of Ver_p. More generally, adjoining ε to Ver_p^+ gives all the even-indexed simples by $\varepsilon \otimes L_j = L_{p-j}$ for odd j. If there are no such intermediate simples, the only possibilities are Vect_k and the subcategory generated by ε. This proves the list.

The multiplication functor in the displayed equivalence sends $\mathbb{1}\boxtimes L_j$ to L_j and $\varepsilon\boxtimes L_j$ to L_{p-j} for odd j. These are all the simple objects of Ver_p, each appearing once. It identifies the corresponding morphism spaces, and is therefore an equivalence of semisimple categories. The symmetry allows the middle two factors of a tensor product to be interchanged, supplying its symmetric monoidal structure. ■

This decomposition and the list of subcategories are also proved in [Ost15, Proposition 3.3]. The product construction and its universal property will be made explicit in the next lecture.

References

[BEO20] Dave Benson, Pavel Etingof, and Victor Ostrik. New incompressible symmetric tensor categories in positive characteristic, 2020.

[CEO23] Kevin Coulembier, Pavel Etingof, and Victor Ostrik. On Frobenius exact symmetric tensor categories. *Annals of Mathematics*, 197(3):1235–1279, 2023.

[CEO24] Kevin Coulembier, Pavel Etingof, and Victor Ostrik. Incompressible tensor categories. *Advances in Mathematics*, 457:109935, 2024.

[CF24] Kevin Coulembier and Johannes Flake. Towards higher Frobenius functors for symmetric tensor categories, 2024. The course discussion follows the May 2024 version.

[Del02] Pierre Deligne. Catégories tensorielles. *Moscow Mathematical Journal*, 2(2):227–248, 2002.

[DM82] Pierre Deligne and James S. Milne. Tannakian categories. In *Hodge Cycles, Motives, and Shimura Varieties*, volume 900 of *Lecture Notes in Mathematics*, pages 101–228. Springer, 1982. Corrected version, 2022, with the original proposition numbering.

[EGNO15] Pavel Etingof, Shlomo Gelaki, Dmitri Nikshych, and Victor Ostrik. *Tensor categories*, volume 205 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, 2015.

[ENO05] Pavel Etingof, Dmitri Nikshych, and Viktor Ostrik. On fusion categories. *Annals of Mathematics*, 162(2):581–642, 2005.

[EO19] Pavel Etingof and Victor Ostrik. On semisimplification of tensor categories, 2019.

[Mil12] James S. Milne. Basic theory of affine group schemes, 2012. Available at www.jmilne.org/math/.

[Ost15] Victor Ostrik. On symmetric fusion categories in positive characteristic, 2015.

[Wat79] William. C. Waterhouse. *Introduction to Affine Group Schemes*. Springer Graduate texts in mathematics. Springer New York, NY, 1979.

[Wib25] Michael Wibmer. A remark on torsors under affine group schemes. *Transformation Groups*, 30:447–454, 2025.